

RESPONSE TO THE SEPTEMBER 25, 2026 REFEREE REPORT

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This response maps each of the 14 major and 25 additional comments to the R40 revision. The central changes are an explicit common-Markov feasibility repair, complete global-search and continuous-state transfer theorems, certified randomized deployments on the unchanged primary cohort, positive nonlinear lower bounds, and a prospectively fixed same-object comparison. Remaining numerical gaps and verification boundaries are stated rather than relabeled as completed results.

1. R40 COMPLETION AND ADDITIONAL SUBSTANTIVE REVISIONS

This version preserves the complete R38 review object and the intervening R39 scientific execution at commit `b5d42b08debd4c26341ab1b1231a6a6242df4f9`. R39 contained new source and checked computations but lacked its final diagnostics and compiled publication. R40 supplies those missing objects, audits their correspondence, and adds the results below. The inherited timed experiments are not presented as newly collected holdout evidence.

1.0.0.1. *Operating approximation: F1, F5, and F7.* The witness-driven repair theorem in the main article gives a globally feasible common policy using U instead of an exact V . Its displayed lattice bracket accounts jointly for witness width, Bellman residual, and policy resolution. The sufficient positive-slack test is computable; a failed test is not a conclusion of economic infeasibility. This addition addresses the operating-oracle objection directly, while preserving every earlier convergence theorem and proof.

1.0.0.2. *Economic content: F9 and additional comments 5, 18, 19, and 25.* A new model-perturbation proposition translates verified parameter uncertainty into operating-tolerance and implementation-cost adjustments for every alternative model, including one policy feasible throughout the uncertainty set. A further cost comparison derives the administration overhead that a certified lottery can absorb while remaining cheaper than every deterministic feasible policy. All 126 initial-law margins, including zeros, are calculated from exact retained endpoints. They are normalized-model economic margins, not empirical estimates or assumed confidence regions.

1.0.0.3. *Independent verification: F4 and F11.* R40 independently replays the entire nonlinear $T = 32$, $N = 128$ object rather than transferring an independent label from the coarser $N = 64$ objects. It also replays all 16 finite proof trees and three explicitly selected continuous-state objects. The complete inherited 42-object primary verification remains hash-linked and separately labeled. The new witness, discontinuity, and local-degeneracy tests are recorded with their actual counts and rejection outcomes.

1.0.0.4. *Complete publication.* Four compiled documents, complete source, generated diagnostics, figures, response mapping, and preservation manifest are supplied on a new branch. The nonlinear $N = 256$ row remains a directed-construction result, not a newly independent scalar replay. Material long-horizon gaps, the tight nonlinear lower-only rows, and the unsolved stopped-control numerical allowance remain visible. No numerical claim is upgraded merely because the publication now compiles.

2. OVERVIEW OF THE REVISION

The report distinguishes a reliable computational record from a complete numerical method. We address that distinction directly. The revised article defines one central object: the minimum integrated cost of a common randomized Markov revision satisfying operating constraints after every state and date. The new theory connects approximate feasibility to that object's optimum, rather than proving only validity of a necessary-action relaxation.

The repair theorem constructs a feasible common continuation and bounds its extra cost by $D_C\delta/[(1-\beta)\varepsilon + \delta]$. It yields a complete finite-state policy-lattice bound and a stopping theorem for the executed interval Bellman search. A separate reset-kernel theorem transfers finite-state intervals to the original measurable-policy problem under explicit total-variation approximation assumptions. These assumptions are not attributed automatically to the atomic maintenance benchmarks.

The computational additions keep the original models and incumbents. Whole-domain randomized policies now serve all restarts and the original uniform initial law. They certify strict cost improvement over every feasible deterministic policy in 27 of the 42 primary configurations. The connected local-support recursion improves 23 primary lower endpoints. In the nonlinear model, the finest horizon-32 interval changes from a zero lower bound to $[8.8718, 11.8492]$. Eight new finite-state environments and both solvers were frozen before generation; every budget stop is retained. The direct McCormick comparison targets exactly the same policy class, initial law, and operating constraints.

The work does not assert that all intervals are now tight. All 30 positive-cost randomized primary intervals retain nonzero gaps. The nonlinear relative gap remains 25.13% at the finest horizon-32 mesh. The stopped diffusion's 0.01 target is not certified, and no synthetic primitive is called calibrated. These statements identify the remaining tasks without replacing the original optimization target or removing the scientific program. All prior article and supplement pages are preserved verbatim; the older historical annex and failed runs remain unchanged.

3. MAJOR COMMENTS

3.1. *R38-F1: Convergence to the original constrained objective*

The revised Section 3 supplies the missing bridge. A policy satisfying the all-restart constraints at $\varepsilon + \delta$ is repaired backward to the original tolerance ε , using one state/date-dependent mixture and one already fixed next-date continuation. The proof establishes measurability, all-restart feasibility, operating-value improvement, and a vanishing implementation-cost modulus. It does not assume strong duality or that outer selectors are feasible.

For finite states, rational policy lattices then give explicit global lower and feasible upper sequences whose gap converges to zero at the actual $K_R(\nu, \varepsilon)$. For the binary-action implementation, the new stopping proposition shows that a surviving box of maximal width w has repaired-midpoint cost within $D_C[w + \rho_\varepsilon(D_J w)]$ of its rectangular Bellman lower bound. Weighted longest-side bisection therefore reaches every positive accuracy without a node or time cap. We report the exponential worst-case depth and do not confuse completeness with practical speed.

The continuous-state transfer in Section 4 additionally identifies sufficient primitive approximation conditions under which finite computation brackets the unrestricted measurable-policy optimum. Thus the paper now has an end-to-end result rather than a convergence statement for an adjacent relaxation. Its assumptions and the atomic-model exception are explicit.

3.2. R38-F2: *The uniform-initial randomized Markov problem*

The revised definition uses one common randomized Markov policy. The primary construction fixes its next-date operating function, compares the installed action with the best action against that continuation, and mixes only enough to restore the current operating inequality. The resulting value is $\max\{q_0, V - \varepsilon\}$. This determines a single measurable policy at every state and date, with fresh conditional randomization rather than a persistent seed. The exact operating verifier checks every knot and open piece.

Costs of these affine-ratio lotteries are enclosed over the whole domain and integrated under the original uniform law. They are paired with valid randomized lower functions for the same object, including the newly connected support recursion. The 42 complete intervals and two nonuniform-law sensitivity tables appear in the supplement. This resolves the policy-class mismatch and establishes 27 strict randomized-versus-deterministic separations. It does not falsely identify the remaining positive-width intervals with exact randomized optima. The finite-state study separately demonstrates complete search on the same common-Markov formulation.

3.3. R38-F3: *Short-horizon concentration and unresolved deterministic cases*

The deterministic results remain intact and are decomposed into 12 zero-cost and 12 positive-cost equalities. Ten positive-cost equalities occur at horizon 4, two at horizon 8, and none at horizon 12. The supplement reports every deterministic relative gap and outer-selector operating violation. We do not improve a headline by counting unchanged zero-cost policies as nontrivial searches.

The new economic result is not a restatement of those deterministic equalities: in 27 cases, a verified randomized upper cost is below the deterministic lower bound. Longer horizons are included in that comparison. The finite stopping theorem also specifies how global search addresses residual constraints rather than assuming tighter operating witnesses will remove a structural outer-relaxation gap.

3.4. R38-F4: *Zero nonlinear lower bounds*

The new nonlinear construction propagates 11 nonnegative global support prices through the entire horizon and combines their positive floors with five local multiplier bounds. Every support is a supersolution for operating reward minus revision cost; its conversion uses the operating lower witness, while local necessary feasibility uses the upper witness. The resulting lower bound is valid for randomized policies and is paired with exactly the retained feasible upper policy at each mesh.

This changes the finest horizon-32 interval to $[8.8718, 11.8492]$, rather than merely certifying feasibility. Complete scalar independent checks cover $(T, N) = (8, 64)$ and $(32, 64)$, including the candidate's upper cost and operating inequalities. The complete $(32, 128)$ object additionally receives a fresh R40 independent replay; the finest $N = 256$ row is not assigned that label. The remaining relative width and memory burden are reported. Four additional $\varepsilon = 0.05$ lower-only diagnostics do not borrow upper policies certified only at 0.5. This is a substantive lower-bound improvement, not a claim that a stringent high-dimensional target has already been solved.

3.5. R38-F5: *One end-to-end error budget*

The reset-approximation theorem gives a single bracket for $K_R(\nu, \varepsilon)$. Certified primitive errors yield operating error a_h and cost error b_h . A relaxed finite problem at $\varepsilon + 2a_h$ gives the lower side; a tightened problem at $\varepsilon - 2a_h$ gives a lifted feasible upper side. The width is

bounded by the two finite optimization widths, $2b_n$, and the repair modulus applied to the $4a_h$ tolerance difference. The measured bracket is a stopping rule for the original objective.

The proof averages arbitrary measurable policies only because the verified reset kernel regenerates a stated within-cell law. It does not assume that a convenient policy sieve is dense in the original feasible set. Atomic shock laws in the primary and nonlinear experiments are explicitly excluded from an automatic vanishing-total-variation claim. Their direct certificates remain valid but their mesh widths are not relabeled as end-to-end approximation errors.

3.6. *R38-F6: Strong same-object baselines*

The direct comparator is a global mathematical-programming formulation in common variables p, J, C , with all Bellman equalities, all restart constraints, and the same initial cost objective. Each policy box uses classical McCormick envelopes. Any proposed LP multiplier is projected to a nonnegative rational vector and accepted only through an exact finite-box residual bound. A floating infeasibility status cannot prune a policy box.

Both methods use the same contraction, midpoint repair, best-bound ordering, split rule, known primitives, node allowance, and wall-clock allowance. The LP method additionally pays for its stronger relaxation and LP-proposed candidate. All eight pairs are reported, including long-horizon failures to close. The comparison often favors the direct relaxation on lower-bound strength; the paper does not assert superiority of interval Bellman search. An uncoupled family of restart occupation measures would relax the common policy constraint, so it is not used as a supposedly identical benchmark.

3.7. *R38-F7: Operating approximation and scaling*

The article now states that exact operating values are inexpensive in the affine one-dimensional development model. Those computations are charged, and they are not evidence about settings where operating approximation is the bottleneck. The nonlinear calculation uses nonzero operating-witness error and nevertheless obtains positive revision lower bounds. Its mesh/horizon table reports full cost intervals, witness diagnostics, payload, and process memory.

The new general theory gives explicit dependence on horizon, discounting, action count, probability resolution, and primitive approximation error. The finite enumeration and spatial-search counts are exponential. The reset theorem displays the state-space mesh dimension burden. These are informative complexity statements, not a dimension-free rate. The current computations do not establish practical high-dimensional certification beyond the reported two-state nonlinear model.

3.8. *R38-F8: Local LP isolated from the main conclusion*

The new primary lower recursion is not the old zero-floor local diagnostic. It propagates nonzero full-horizon support floors at every date, evaluates local pure and two-action boundary vertices with directed cell data, and is paired with the feasible common-Markov lottery under the same initial law. It improves 23 of the primary uniform lower endpoints. This directly changes the reported global interval.

The inherited adaptive local-LP theorem and all 12 executions remain available, with interval-width-versus-cell and interval-width-versus-time figures. Their local scope is not enlarged retroactively. The new fixed-partition vertex calculation requires no generic nondegeneracy assumption: boundary pure actions are included and equal operating coefficients do not trigger division. Finite support prices still need not close a global duality gap.

3.9. *R38-F9: Economic content and journal relevance*

The revision strengthens the general numerical-methods route rather than attaching an unsupported calibration to the synthetic model. The new theorem pair supplies completeness for the actual constrained common-policy objective and an explicit transfer/error budget under verifiable approximation assumptions. The primary economic comparison identifies a specific effect: admissible randomized action assignment can strictly reduce implementation cost relative to every deterministic feasible rule, when cost is paid on realized departures.

The paper does not infer monetary welfare, empirical prevalence, or estimated adjustment costs from neural labels or designed thresholds. Fixed randomization overhead and parameter uncertainty would change the primitives and would need their own certificates. The joint-perturbation study improves evaluation design but is not an empirical application. The economic interpretation is therefore tied to a precise proved policy-class comparison, not to success percentages on hand-designed rules.

3.10. *R38-F10: Novelty and literature positioning*

The literature discussion now distinguishes constrained occupation-measure formulations, finite approximations to constrained control, safe/primal–dual policy methods, global nonconvex programming, semi-infinite optimization, verified numerical analysis, and economic replacement models. The bibliography expands from ten to eighteen entries, retaining all earlier references. McCormick convexification, weak duality, Bellman comparison, and rational verification are explicitly treated as established ingredients.

The theorem-level advance claimed is the common-continuation repair and its explicit tolerance-to-cost modulus, combined with complete search and a reset-kernel transfer for all-restart constraints. The direct global-programming comparator demonstrates what a standard formulation returns on the identical finite objective and exposes the cost of stronger relaxations. The article does not claim that repository organization alone constitutes scientific novelty or that existing global optimization cannot solve the same problem.

3.11. *R38-F11: Independent verification and model contracts*

The new finite verifier imports neither solver nor LP optimizer. It reconstructs the economic stage contract, common-policy equations, all necessary contractions, branch coverage, every accepted LP residual bound, and the final leaf minimum for all 16 trees. The continuum verifier independently checks 42 complete lottery objects, their cost enclosures, nonzero support families, and connected local vertices using SymPy arithmetic rather than construction operations. The nonlinear verifier independently rebuilds rational box geometry and scalar recursions for two complete representative objects.

The summary tables distinguish these checks from same-code structural validation, cross-run identity, and publication-integrity hashes. Inherited bounds keep their original provenance; taking their maximum with a new independent bound does not make every inherited component independently reverified in R39. Seeded adversarial partition tests and deliberate mutation rejection are supplementary tests, not proofs of a false-acceptance probability. Analytic assumptions such as the reset-kernel approximation contract remain mathematical obligations rather than machine-verified properties of every model.

3.12. *R38-F12: Stopped-control material*

The continuous-action stopped consumption–portfolio target remains in the preserved scientific record, including its derivations and the inherited bound near 7.1818 against a 0.01

target. The current main article does not use a killed-operator analogy as numerical validation of that target. It explains the outstanding action, boundary, quadrature, time-discretization, and diffusion-approximation requirements in the context of the new error budget.

No derivation or prior result is deleted. The complete prior article and supplement are appended verbatim, while the older historical annex remains archival rather than part of the current article’s proof burden. This preserves the program without substituting unsupported applicability for a verified numerical result.

3.13. *R38-F13: Coherence of the article*

The organizing object is now $K_R(\nu, \varepsilon)$ for a common randomized Markov rule. Sections 2–4 define it, prove repair and convergence, and supply the continuous-model transfer. Section 5 explains which inherited certificates accelerate or bound that object and which retain a different policy-class scope. Section 6 evaluates the common-policy solvers, whole-domain primary deployments, and nonlinear lower bounds. Verification and economic interpretation follow that same structure.

The historical frontier, analytic two-period example, sensitivity study, and stopped diffusion are retained but are not conflated with the common-policy convergence theorem. A fixed-restart history-conditioned solution is still labeled as such. The current main result can be read without reconstructing the older 349-page archive.

3.14. *R38-F14: Development versus evaluation*

The 42 primary configurations are explicitly called a development cohort. Their frozen incumbent files preserve identity, not out-of-sample status. The new eight-environment protocol, solver hashes, case seeds, joint perturbations, and fixed budgets were committed before generating those environments. Both methods were executed on every case, and every remaining interval was retained. The publication identifies later CI execution as a replay, not a newly untouched test.

The new suite varies finite state count, horizon, discounting, operating allowance, intervention cost, transition interactions, and initial masses. It is still a designed finite-state test, not a broad statistical sample or an evaluation of all possible continuous models. This distinction is stated beside the numerical results rather than only in a repository note.

4. ADDITIONAL TECHNICAL AND PRESENTATION COMMENTS

4.1. *1. Relative gaps*

All main numerical tables now display gap divided by a positive feasible upper cost. Full supplement tables retain absolute gaps as well. Exact zero upper costs are separated rather than assigned a misleading relative accuracy.

4.2. *2. Zero-cost exactness*

The article explicitly separates the 12 zero-cost primary randomized cases from the 30 positive-cost cases and explains that zero-cost optimality follows from feasibility and nonnegativity. The abstract’s main numerical statement is strict randomized cost improvement, not a combined exactness count.

4.3. 3. *Horizon distribution of deterministic exactness*

The primary-results subsection states the ten/two/zero split of positive deterministic equalities at horizons 4/8/12. Since deterministic equality is no longer the abstract's headline, we do not repeat its full breakdown there as if it were the main common-randomized result. It is immediately available in the results and response.

4.4. 4. *Meaning of global bounds*

Every bound names its policy class and objective. Necessary-action deterministic lower functions are not labeled randomized lower functions. The primary randomized interval uses the same dispersed initial law on both sides, while fixed-restart history-conditioned results retain their separate scope.

4.5. 5. *Uniform initial law and sensitivity*

Uniform initialization is a transparent integration convention, not an empirical distribution. The primary calculation now includes complete sensitivity tables for densities $2x$ and $2(1 - x)$ under the same common policies and operating constraints. They yield 28 and 27 strict deterministic separations, respectively; these are not independent observations.

4.6. 6. *The exceptional horizon-eight equality*

The new exact diagnostic refines the earlier wording: $C_0 = B_0$ pointwise, including knots, in the neural31003 case at allowance $1/20$. The failure of the full all-date equality flag arises at dates 1, 2, and 3. The supplement lists every nonzero affine interval and isolated point with exact rational endpoints. Positive-gap regions have zero probability under the selected deployment from the uniform initial law but matter for alternative restarts. The difference is not described as an isolated initial-state artifact.

4.7. 7. *Remaining gaps and installed-policy normalization*

The supplement reports all 42 deterministic gaps, gap/upper, selector deficit, and excess deficit over tolerance, as well as all randomized primary intervals. The installed rule has zero revision cost by definition, making a ratio to its revision cost undefined. We state this explicitly instead of reporting zero or substituting an operating payoff with different units.

4.8. 8. *Assumption table*

The supplement's opening table maps each theorem and experiment to policy class, initial law, dimension, arithmetic, bound sources, and independent coverage. It separately identifies analytic approximation assumptions and checked finite proof objects.

4.9. 9. *Borel measurability*

The main deterministic-recursion subsection proves that the necessary correspondence has a Borel graph as a finite union of sublevel sets. Backward lower functions and lowest-index minimizing selectors are Borel. The repair proof likewise states its Borel mixing set and avoids division on the zero-denominator set.

4.10. 10. Attainment versus infimum

The general Borel problem is written as an infimum. Its repair value modulus uses arbitrarily near-minimizing policies. Finite-state common-policy problems attain minima by compactness and continuity, and finite represented enumerations attain them by finiteness. No general function-space attainment is assumed.

4.11. 11. Degeneracies and termination

The new local vertex method handles equality and coincident operating coefficients explicitly without dividing by a zero difference. Its fixed partition does not invoke the inherited adaptive nondegeneracy assumption. The old adaptive logs do not contain a complete degeneracy occurrence count; we do not manufacture one from successful runs. Their conditional theorem and records remain intact.

4.12. 12. Error versus work curves

Two figures in the supplement plot all 12 inherited local intervals against verified cells and construction seconds. The full numerical table is retained, exposing rather than hiding the sharp work growth. These plots describe the local relaxation, not global common-policy optimality.

4.13. 13. Why nonlinear lower bounds were zero

The new full-horizon support construction yields positive bounds, showing that zero was not structurally forced by the definition of revision cost. The table separates the support-only and connected restart lower endpoints. Remaining coarseness and operating-witness error still affect strength.

4.14. 14. Matched nonlinear comparisons

The revision makes no new upper-portfolio success-count comparison. It pairs the stronger lower bound with exactly the retained feasible upper candidates at identical horizon, mesh, and tolerance. A genuine matched global-solver comparison is instead performed on the frozen finite-state suite, with identical node and wall-clock allowances.

4.15. 15. Memory scaling

The main nonlinear subsection gives $8N^2(T + 1)(11 + 4)$ bytes for retained support/witness arrays before temporary storage. The supplement gives array MiB by mesh and horizon; machine records also report process peak resident memory.

4.16. 16. Uncertified versus infeasible

A failed sufficient nonlinear box check is labeled uncertified. The finite search removes boxes only through necessary infeasibility or valid cost bounds. The one-dimensional selector can be called infeasible only where its exact operating calculation exhibits a violation.

4.17. 17. Costs on threshold-crossing boxes

The supplement explains that upper revision cost on such a box can be a conservative enclosure: the installed action may vary within the box even when the implemented action is fixed. This does not imply that every realized state changes action. Lower and upper enclosure conventions are separately checked.

4.18. 18. *Joint perturbations*

The new prospective finite suite changes persistence, interaction terms, intervention effect and price, discounting, horizon, tolerance, and initial distribution jointly. The old one-at-a-time sensitivities remain preserved but are no longer treated as the sole robustness evidence.

4.19. 19. *Economic information in neural incumbents*

Neural and spline incumbents are explicitly fixed computational policies, not estimated economic objects or evidence about preferences. The new economic conclusion depends on realized revision cost and admissible randomization, not on the architecture label of the installed rule.

4.20. 20. *Reproducible environment*

The computation report records exact Python and package versions, CPU and operating-system information, thread limits, source hashes, and wheel hashes in the CI logs. The workflow provides an executable environment recipe. Hosted runner image labels are not claimed to be permanent container digests; long-run binary identity is distinguished from logical rational-certificate verification.

4.21. 21. *Historical annex burden*

The main article supplies its definitions and proofs without requiring the historical annex. Prior scientific material is nevertheless preserved verbatim in the appended R38 documents and unchanged archive. Preservation is not presented as part of the new numerical evidence.

4.22. 22. *Verification terminology*

Independent arithmetic, same-code read-only checks, cross-run identity, and publication integrity are separately described in the assumption and coverage tables. Representative nonlinear checks do not confer independent status on all finer meshes. Historical lower components retain their provenance.

4.23. 23. *Property-based adversarial tests*

The new seeded suite tests 200 randomly generated discontinuous piecewise-affine pairs, their lower envelopes and pullbacks, and 400 malformed partitions, in addition to nine finite-tree mutation categories. Exact comparisons use a separate SymPy evaluator. The report states coverage and limitations rather than interpreting mutation rejection as an all-input correctness probability.

4.24. 24. *What standard formulations return*

The literature section and direct comparator explain the common-policy compatibility constraint. A classical nonconvex Bellman formulation with McCormick spatial bounds returns certified intervals on the identical finite objective, and its actual outcomes are reported. We do not claim that standard constrained-control formulations are incapable of doing so or that our interval recursion dominates them.

4.25. 25. *External validity*

The conclusion ties the economic statement to certified policy-class comparisons and the general theorem to explicit assumptions. The synthetic sensitivity and nonlinear sections remain designed numerical diagnostics, not empirical prevalence or broad external-validity estimates.

5. DOCUMENTS SUPPLIED FOR RE-REVIEW

The branch supplies the revised article, this point-by-point response, the technical supplement with full tables and preserved R38 documents, and a computation report with executable commands and provenance. New source, exact proof objects, directed nonlinear arrays, all stop records, independent reports, and publication checks are retained. The revision changes the method and the evidence, while leaving remaining numerical and application gaps visible for the next review.